WEEK 11 / LESSON 2 / 50 MINUTES

L2

The lab bench of ML learning

Reproduction is the weightlifting of ML. It exposes silent assumptions — the hyperparameter the author forgot, the seed that changed behaviour, the figure that was cherry-picked. It is the fastest way to become respected in AI.

Which paper do you wish you had reproduced last year?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

You can read 100 papers and not understand one. Reproduce 1 and everything clicks.

WHY
Real
A real reason this matters in industry today.

SCALE
Today
A real reason this matters in industry today.

SPEED
Now
A real reason this matters in industry today.

IMPACT
Yours
A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. 5 reproducibility signals.
2. Build a reproduction scaffold.
3. Debug 5 common failures.
4. Decide when to stop.
5. A shareable reproduction repo.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

5 reproducibility signals

Key concept of this lesson.

02

Reproduction scaffold

Boring wins

03

5 common failures

Key concept of this lesson.

04

When to stop

Senior discipline

01

PART 1 OF 4

5 reproducibility signals

Key concept of this lesson.

CONCEPT 1 OF 4

5 reproducibility signals

A core idea you'll use throughout this lesson.

DEFINITION

5 reproducibility signals

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Reproduction scaffold

Fork + pin commit hash.
Docker / Colab environment.
Set ONE seed across torch, numpy, random.
Run the smallest config first.
Log every number to W&B or CSV.
SETUP
Boring

Reproducibility is discipline.

KEY POINTS

- Fork + pin commit hash
- Docker / Colab environment
- Set ONE seed across torch, numpy, random
- Run the smallest config first

CONCEPT 3 OF 4

5 common failures

A core idea you'll use throughout this lesson.

DEFINITION

5 common failures

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

When to stop

Within 5% of reported number → stop + publish.
Stuck 3 days on environment → ask + move on.
Paper irreducibly missing → file an issue, pivot.
Out-of-compute → reproduce at smaller scale + disclose.
STOP
5% rule

Perfection kills publication.

KEY POINTS

- Within 5% of reported number → stop + publish
- Stuck 3 days on environment → ask + move on
- Paper irreducibly missing → file an issue, pivot
- Out-of-compute → reproduce at smaller scale + disclose

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Build a repro scaffold

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.
2.
3.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Which reproducibility signal do your own projects fail on?*

Q2 *What deviation are you afraid to disclose?*

Q3 *Which paper could you reproduce in 2 hours?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Biggest reproducibility signal:

A Twitter followers

B Code released

C Press coverage

D Poster

ANSWER · QUESTION 1 OF 3

Biggest reproducibility signal:

B

CORRECT ANSWER

Code released

WHY

Code is decisive

KNOWLEDGE CHECK · QUESTION 2 OF 3

Seed count for credible results:

A B

B ≥ 3

C E

D (no option)

ANSWER · QUESTION 2 OF 3

Seed count for credible results:

C

CORRECT ANSWER

E

WHY

≥ 3 seeds → mean + std

KNOWLEDGE CHECK · QUESTION 3 OF 3

Accept gap:

- **A** 0%
- **B** 5%
- **C** 50%
- **D** 100%

ANSWER · QUESTION 3 OF 3

Accept gap:

CORRECT ANSWER

B

5%

WHY

Within 5% → stop + publish

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

"The reproducers become the reviewers become the researchers."

WHAT TO NOTICE

1. 5 reproducibility signals
2. Reproduction scaffold
3. 5 common failures
4. When to stop

APPLY IT → ASSIGNMENT

Plan your reproduction

Pick a paper for Saturday's P1.

DELIVERABLES

- Score 5 signals.
- Draft a 2-hour scaffold.
- Share pick on LMS forum.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Repo URL by Sat.

ONE THING TO REMEMBER

“

"The reproducers become the reviewers become the researchers."

— AI Learning Hub

END OF LESSON 11

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 3 — Landmark Papers.